Practical variant effect prediction with deep mutational scanning data

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Galaxy PROJECT

GHGA · ELIXIR-DE · de.NBI
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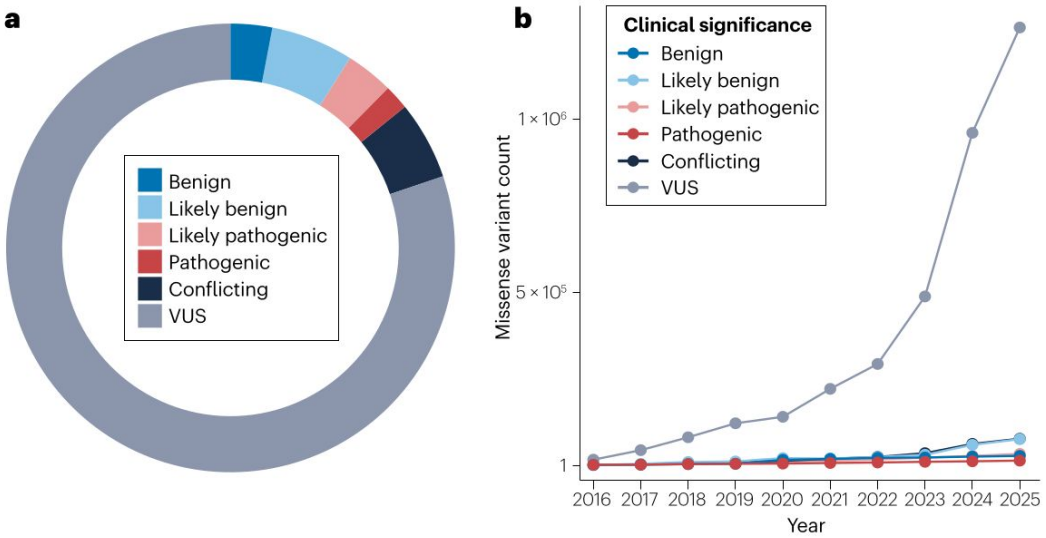
Summary

1 INTRO. The variant interpretation gap

- Millions of genomes sequenced
- Hundreds of millions of missense variants identified
- Majority classified as VUS
- Functional evidence is limited
- Computational predictors (SIFT, PolyPhen, EVE, etc.) often disagree
- Clinical decisions require stronger evidence

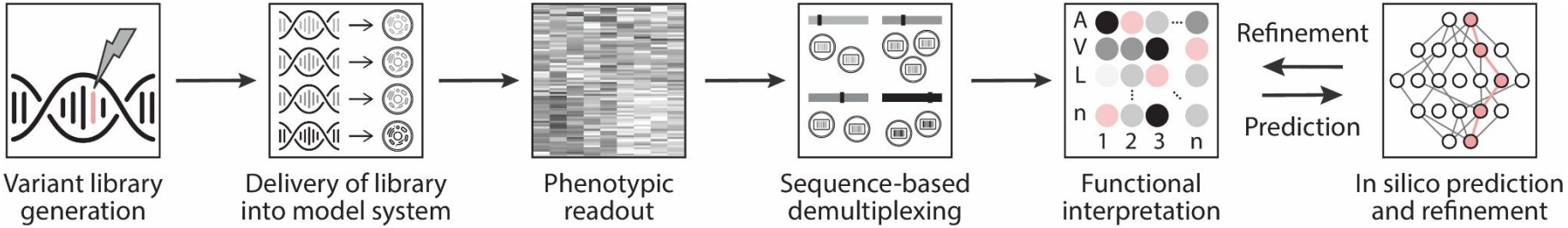


Source: McEwen et al., 2025 (fig. 1)

- Motivation for scalable functional assays

1 INTRO. Sequence-function approach

- Traditional functional studies: slow, variant-by-variant
- Sequence-function studies enable:
  - Systematic testing
  - Thousands of variants in parallel
  - Quantitative effect maps
- Toward a Variant Effect Atlas

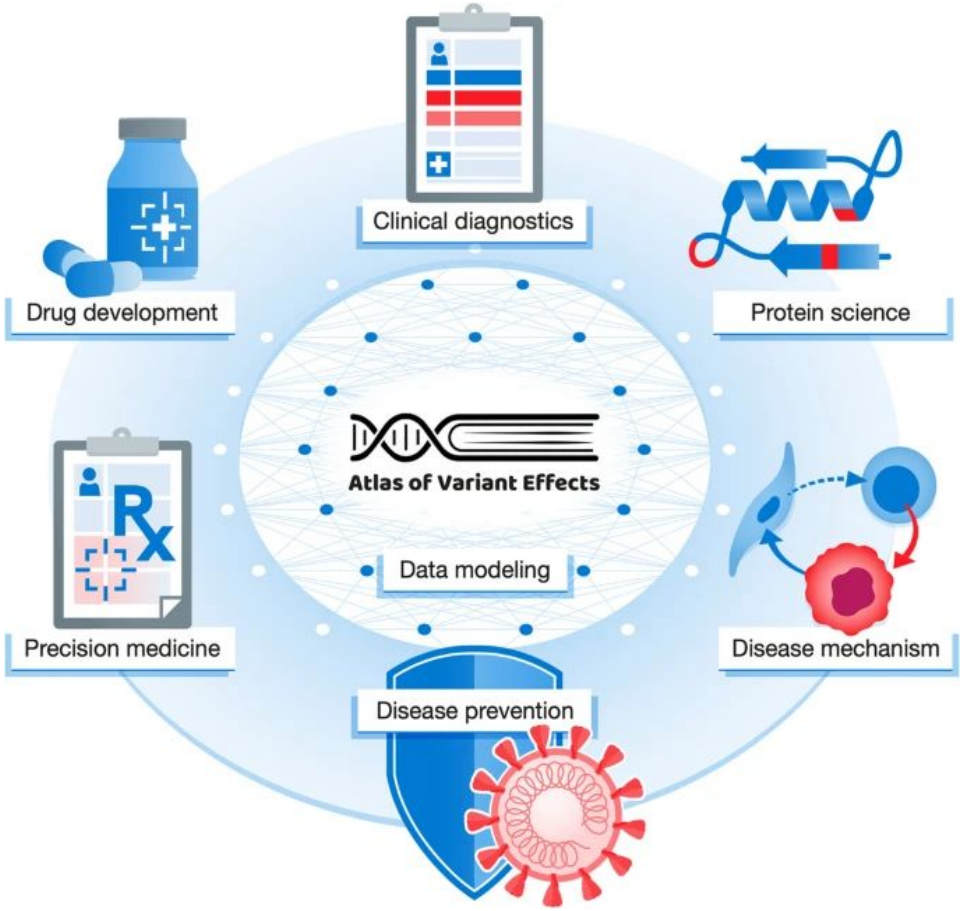


Tabet D, et al. 2022
Annu. Rev. Genet. 56:441–65

Source: Tabet et al., 2022 (fig. 2)

2 INTRO. Significance

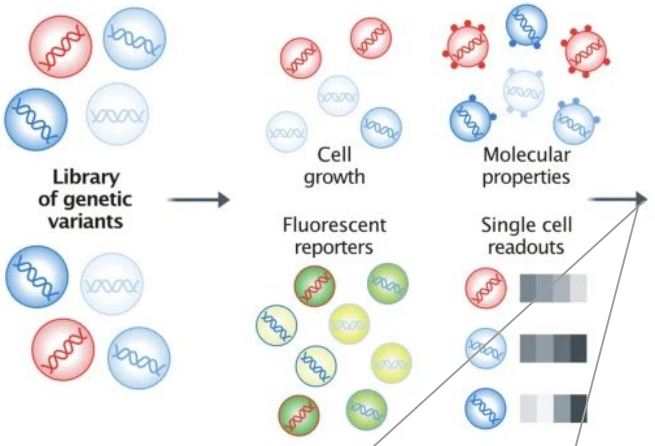
- Sequence-function studies help to understand genetic variants in diseases like cancer.
- Supports precision medicine and therapeutic development, etc.



Source: Fowler et al., 2023 (fig. 1)

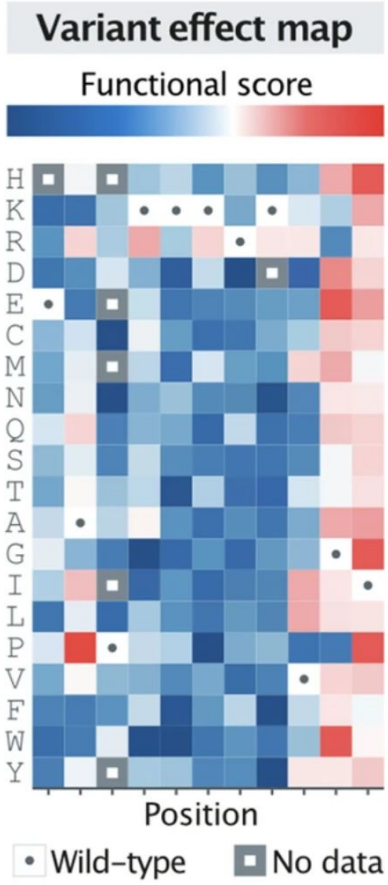
2 DMS. Deep Mutational Scanning

- High-throughput experimental assay
- Systematically tests thousands of variants
- Measures functional effect (fitness, binding, stability, etc.)
- Produces quantitative variant effect maps
- DMS score reflects performance in a specific assay - not clinical pathogenicity!



Source: Fowler et al., 2023 (fig. 2A)

Source: https://github.com/rothlab/tileseqMave



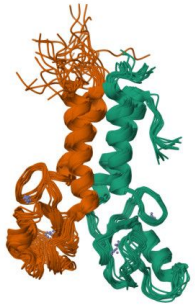
2 DMS. What does DMS score mean?

- Relative to wild-type
- Assay-specific phenotype
- Often normalized, e.g.,
  - synonymous = neutral
  - nonsense = loss-of-function
- Continuous scale
  - [0, 1] (DMS-TileSeq)
  - (-∞, 0] (Enrich2)
  - [-1,1]
  - etc.

2 DMS. Types of DMS assays

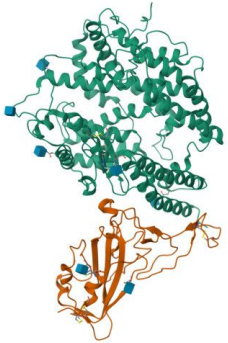
Protein stability

- Measures intracellular abundance
- fluorescence reflects protein abundance
- e.g., BRCA1 RING domain



Binding

- Measures protein-protein interaction
- fluorescence reflects binding level
- e.g., SARS-CoV-2 RBD-ACE2



Enzymatic activity

- Measures catalytic function
- activity coupled to bacterial growth under antibiotic pressure
- e.g., TEM-1 beta-lactamase

Cellular fitness

- Measures growth under stress
- growth competition measured by enrichment
- e.g., CHK2 (yeast complementation under DNA damage)

2 DMS. Strengths & Limitations

Strengths

- Empirical functional data
- Single-residue resolution
- Quantitative
- Scalable

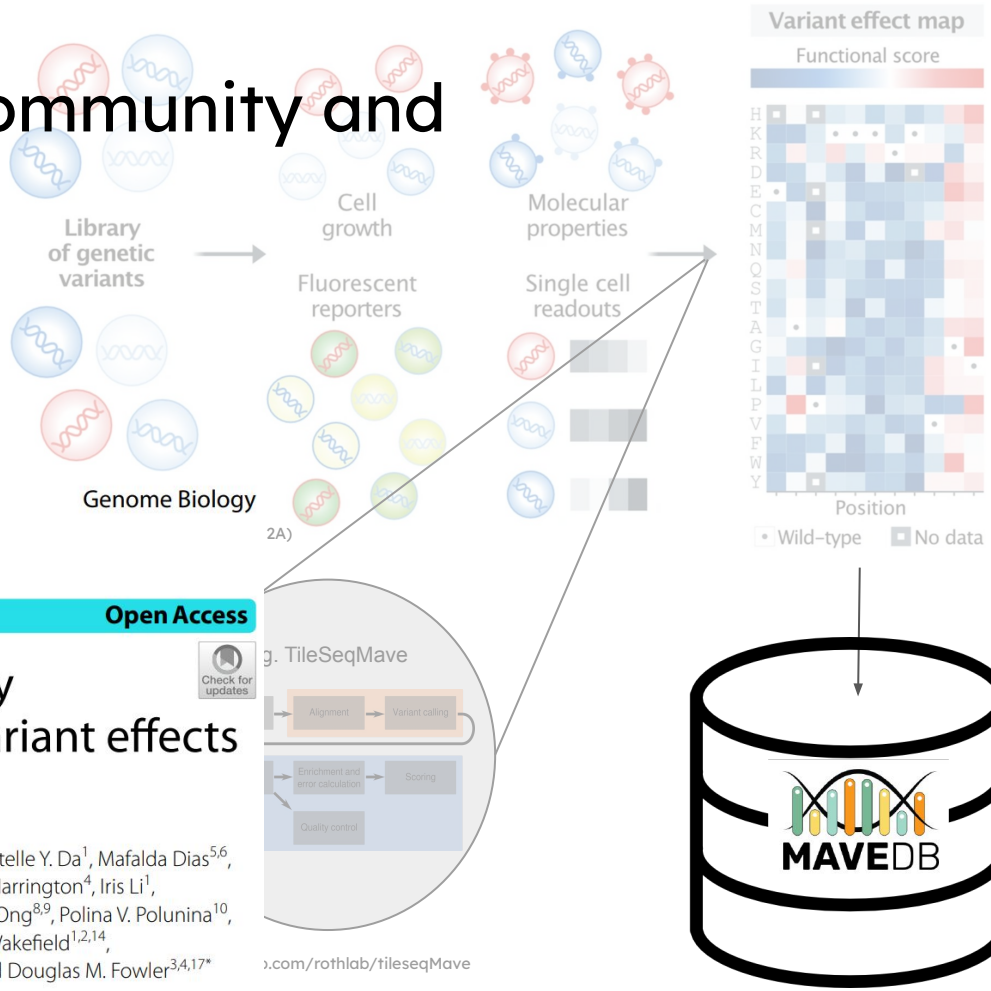
Limitations

- Assay-specific
- Context-dependent
- Incomplete coverage
- Different score scales

3 MAVEDB. The MAVE community and MaveDB

- Central repository for DMS/MAVE data.
- Standardized format
- Score sets with metadata
- Enables reuse & reproducibility

Rubin *et al. Genome Biology* (2025) 26:13
https://doi.org/10.1186/s13059-025-03476-y



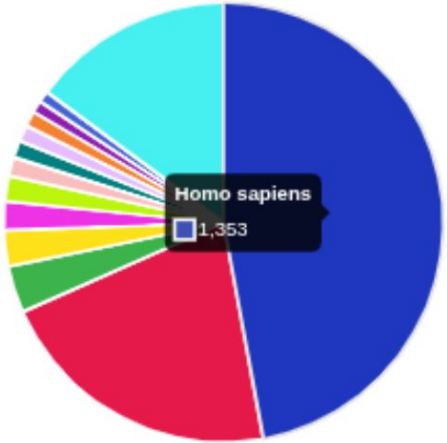
DATABASE **Open Access**

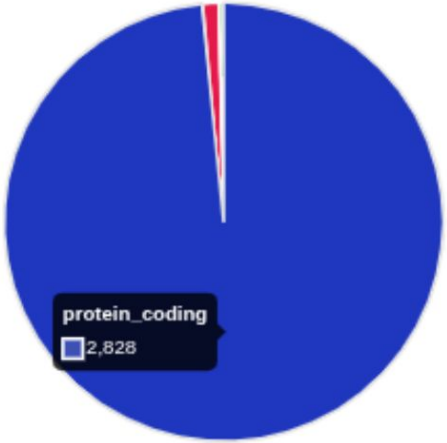
MaveDB 2024: a curated community database with over seven million variant effects from multiplexed functional assays

Alan F. Rubin[1,2*], Jeremy Stone[3], Aisha Haley Bianchi[4], Benjamin J. Capodanno[3], Estelle Y. Da[1], Mafalda Dias[5,6], Daniel Esposito[1], Jonathan Frazer[5,6], Yunfan Fu[1,2], Sally B. Grindstaff[3], Matthew R. Harrington[4], Iris Li[1], Abbye E. McEwen[3,4,7], Joseph K. Min[4], Nick Moore[1], Olivia G. Moscatelli[2,8], Jesslyn Ong[8,9], Polina V. Polunina[10], Joshua E. Rollins[11], Nathan J. Rollins[12], Ashley E. Snyder[3], Amy Tam[13], Matthew J. Wakefield[1,2,14], Shenyi Sunny Ye[4], Lea M. Starita[3,4], Vanessa L. Bryant[2,8,15], Debora S. Marks[13,16*] and Douglas M. Fowler[3,4,17*]

3 MAVEDB. Scale

- > 2600 datasets
- Millions of variant scores
- ~ 100 target organisms
- Coding + noncoding





Top-10 target organisms in MaveDB	Associated score sets
Homo sapiens	1352
artificial sequences	611
Gallus gallus	99
Streptococcus sp. 'group G'	76
Saccharomyces cerevisiae S288C	60
Zoarces americanus	54
Escherichia coli K-12	40
Severe acute respiratory syndrome coronavirus 2	36
Geobacillus stearothermophilus	35
Mus musculus	32

3 MAVEDB. MaveMD - clinical interface for DMS/MAVE data

- Built on top of MaveDB
- Designed for clinical variant interpretation
- Integrates:
  - ClinVar pathogenic / benign variants
  - ClinGen Allele Registry
  - Evidence calibration (ACMG/AMP PS3/BS3)
- Displays:
  - Assay facts (model system, phenotype measured)
  - Clinical performance
  - Evidence strength (OddsPath, ExCALIBR)

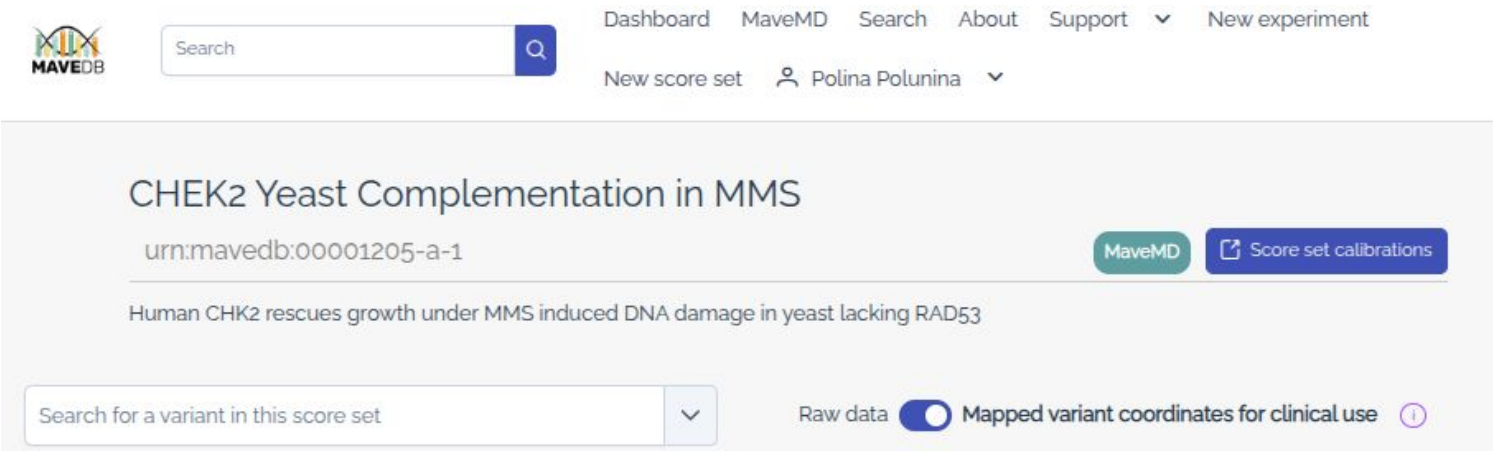
Preprint

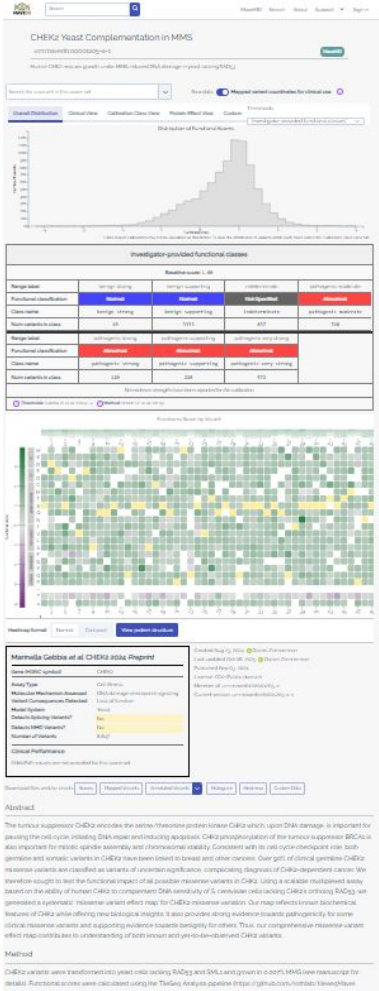
MaveMD:A functional data resource for genomic medicine

Abbye E. McEwen, Jeremy Stone, Malvika Tejura, Pankhuri Gupta, Benjamin J. Capodanno, Estelle Y. Da, Sally B. Grindstaff, Nick Moore, David Reinhart, Ashley E. Snyder, Andrew B. Stergachis, Lea M. Starita, Douglas M. Fowler, Alan F. Rubin

doi: https://doi.org/10.1101/2025.11.15.25336228

3 MAVEDB. Example dataset (CHK2)





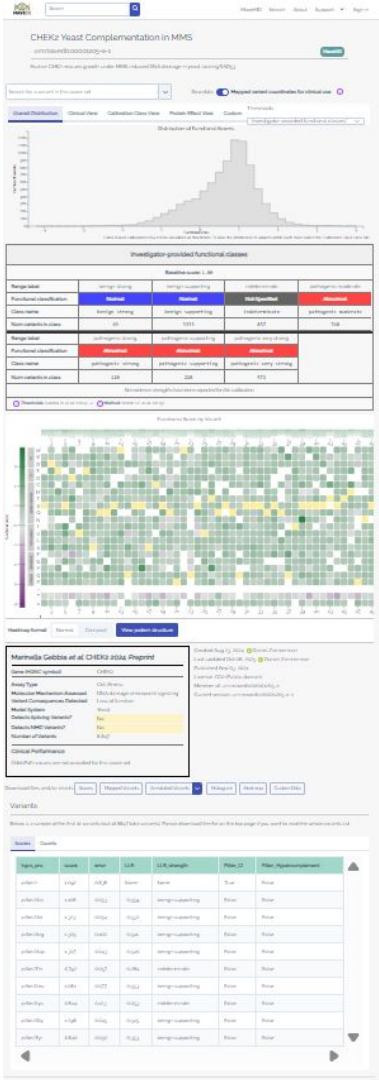
3 MAVEDB. CHK2: variant-level scores

Variants

Below is a sample of the first 10 variants (out of 8847 total variants). Please download the file on the top page if you want to read the whole variants list.

Scores | Counts

hgvs_pro	score	error	LLR	LLR_strength	Filter_CI	Filter_Hypercomplement
p.Ser2=	1.092	0.836	None	None	True	False
p.Ser2Ala	1.106	0.053	-0.559	benign supporting	False	False
p.Ser2Val	1.323	0.054	-0.532	benign supporting	False	False
p.Ser2Arg	1.305	0.100	-0.541	benign supporting	False	False
p.Ser2Asp	1.307	0.043	-0.540	benign supporting	False	False
p.Ser2Thr	0.792	0.057	-0.284	indeterminate	False	False
p.Ser2Leu	1.081	0.077	-0.553	benign supporting	False	False
p.Ser2Lys	0.644	0.123	-0.053	indeterminate	False	False
p.Ser2Gly	1.298	0.045	-0.545	benign supporting	False	False
p.Ser2Tyr	0.840	0.050	-0.353	benign supporting	False	False



3 MAVEDB. CHK2: assay metadata

Marinella Gebbia *et al.* CHEK2 2024 *Preprint*

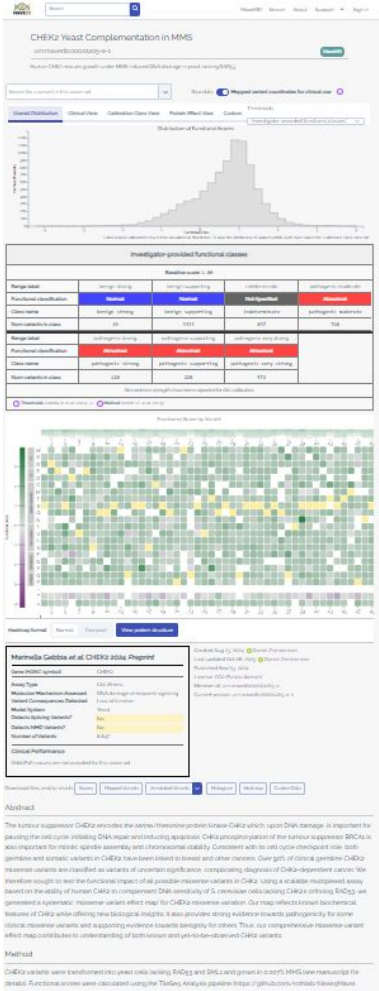
Gene (HGNC symbol)	CHEK2
Assay Type	Cell fitness
Molecular Mechanism Assessed	DNA damage checkpoint signaling
Variant Consequences Detected	Loss of function
Model System	Yeast
Detects Splicing Variants?	No
Detects NMD Variants?	No
Number of Variants	8,847

Clinical Performance

OddsPath values are not provided for this score set.

Created Aug 23, 2024 Daniel Zimmerman
Last updated Oct 06, 2025 Daniel Zimmerman
Published Sep 03, 2024
License: CC0 (Public domain)
Member of: urn:mavedb:00001205-a
Current version: urn:mavedb:00001205-a-1

Download files and/or charts: Scores | Mapped Variants | Annotated Variants | Histogram | Heatmap | Custom Data



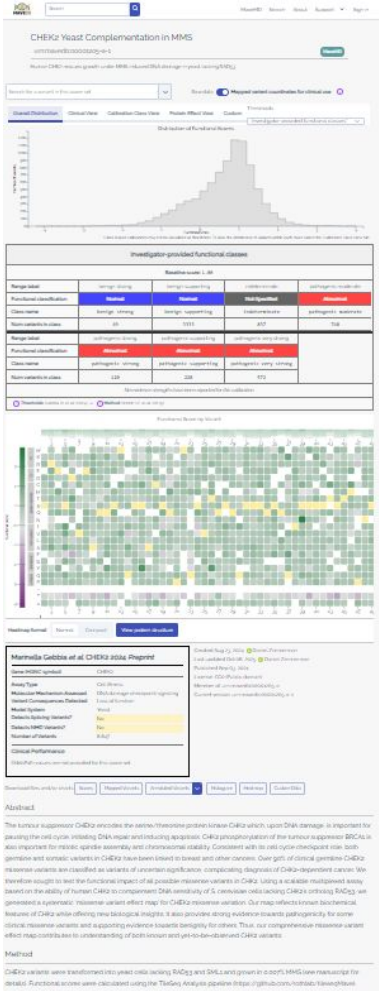
3 MAVEDB. CHK2: study description

Abstract

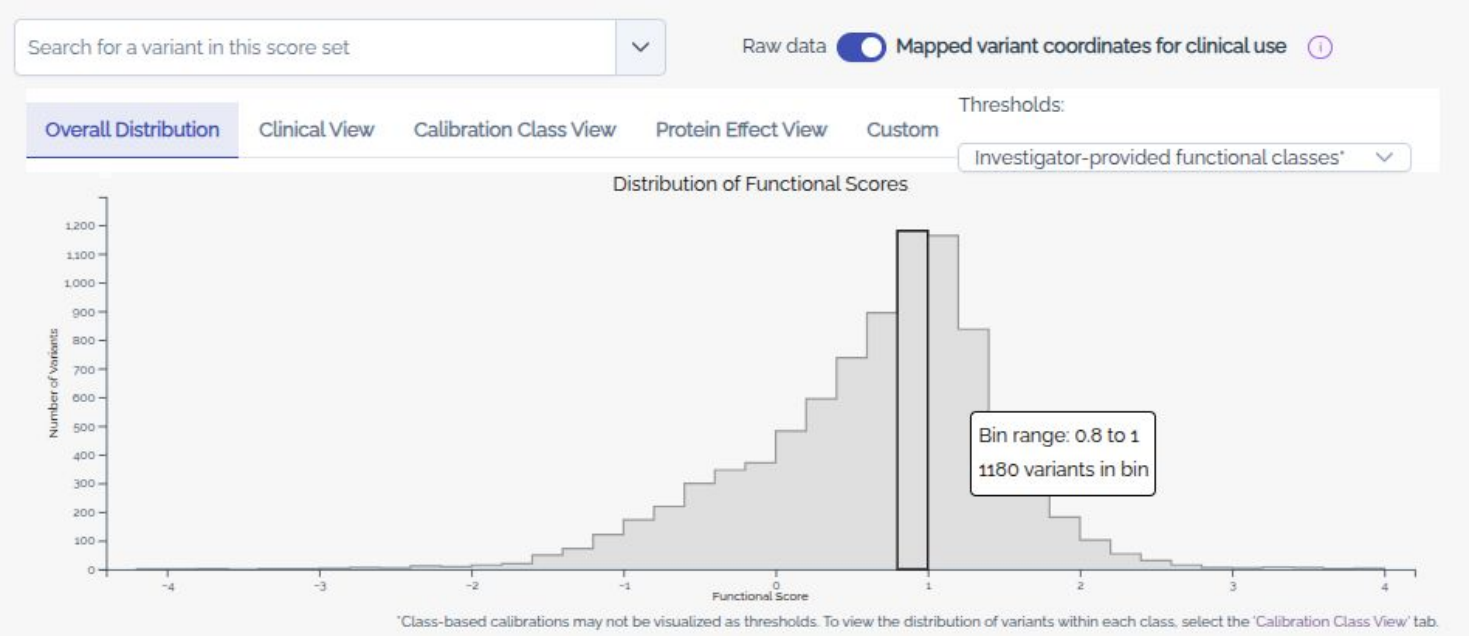
The tumour suppressor CHEK2 encodes the serine/threonine protein kinase CHK2 which, upon DNA damage, is important for pausing the cell cycle, initiating DNA repair and inducing apoptosis. CHK2 phosphorylation of the tumour suppressor BRCA1 is also important for mitotic spindle assembly and chromosomal stability. Consistent with its cell cycle checkpoint role, both germline and somatic variants in CHEK2 have been linked to breast and other cancers. Over 90% of clinical germline CHEK2 missense variants are classified as variants of uncertain significance, complicating diagnosis of CHK2-dependent cancer. We therefore sought to test the functional impact of all possible missense variants in CHK2. Using a scalable multiplexed assay based on the ability of human CHK2 to complement DNA sensitivity of S. cerevisiae cells lacking CHK2's ortholog RAD53, we generated a systematic 'missense variant effect map' for CHEK2 missense variation. Our map reflects known biochemical features of CHK2 while offering new biological insights. It also provides strong evidence towards pathogenicity for some clinical missense variants and supporting evidence towards benignity for others. Thus, our comprehensive missense variant effect map contributes to understanding of both known and yet-to-be-observed CHK2 variants.

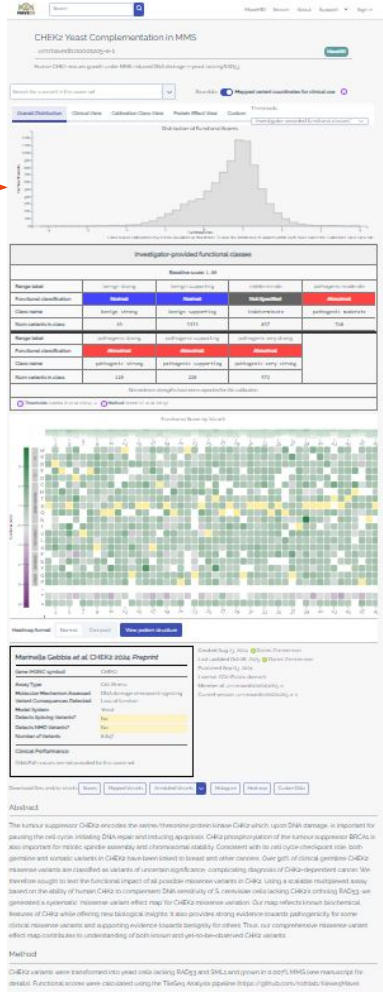
Method

CHEK2 variants were transformed into yeast cells lacking RAD53 and SML1 and grown in 0.007% MMS (see manuscript for details). Functional scores were calculated using the TileSeq Analysis pipeline (https://github.com/rothlab/tileseqMave).

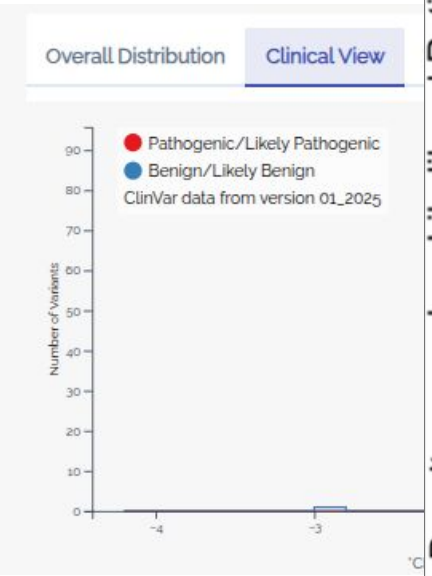


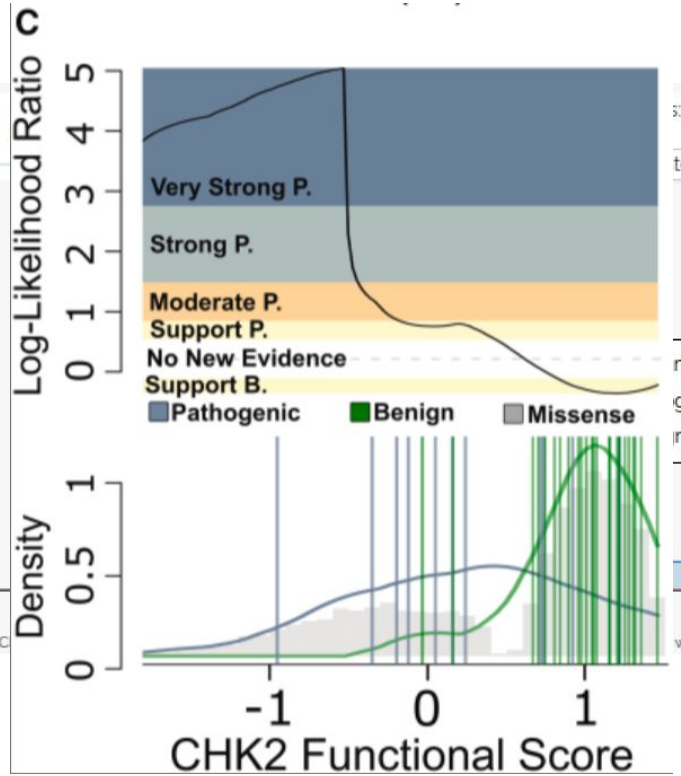
3 MAVEDB. CHK2: distribution of scores

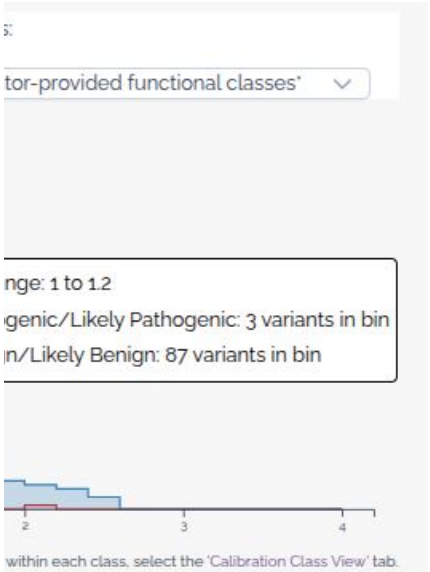


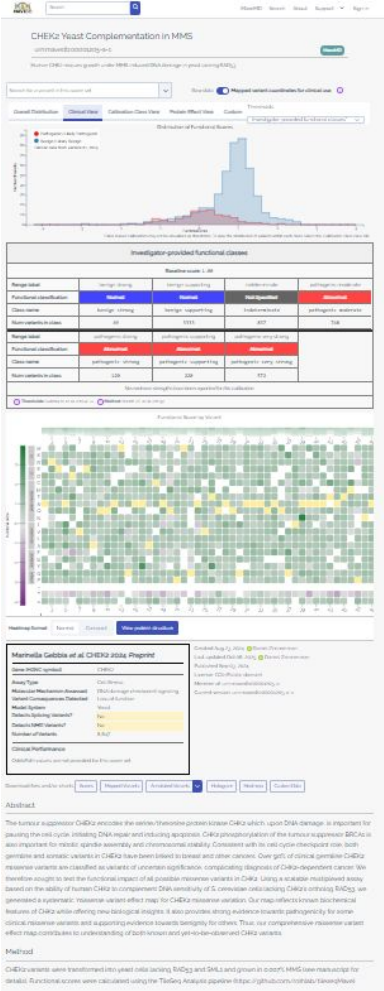


3 MAVEDB. CHK2: ClinVar classifications







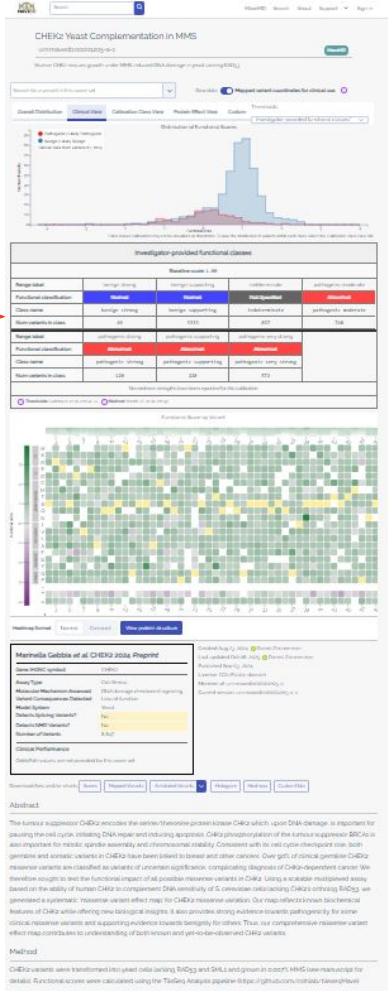


Source: Gebbia et al., 2024 (fig. 7C)

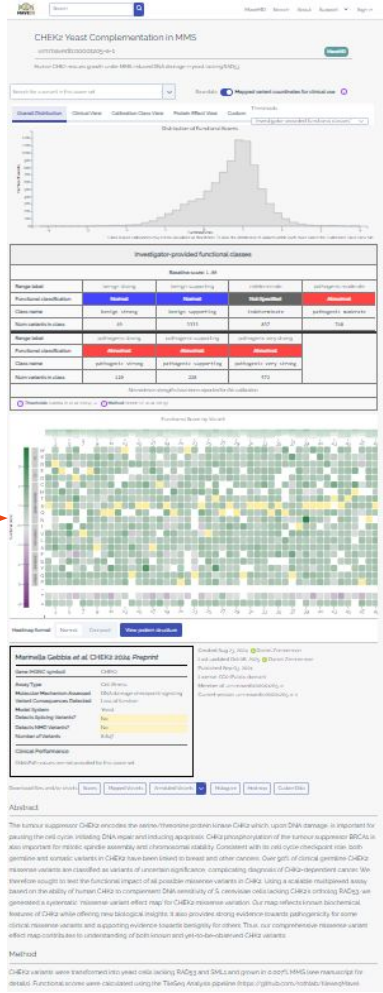
3 MAVEDB. CHK2: calibration classes

Investigator-provided functional classes				
Baseline score: 1.00				
Range label	benign strong	benign supporting	indeterminate	pathogenic moderate
Functional classification	Normal	Normal	Not Specified	Abnormal
Class name	benign strong	benign supporting	indeterminate	pathogenic moderate
Num variants in class	49	3333	837	748
Range label	pathogenic strong	pathogenic supporting	pathogenic very strong	
Functional classification	Abnormal	Abnormal	Abnormal	
Class name	pathogenic strong	pathogenic supporting	pathogenic very strong	
Num variants in class	120	220	572	
No evidence strengths have been reported for this calibration.				

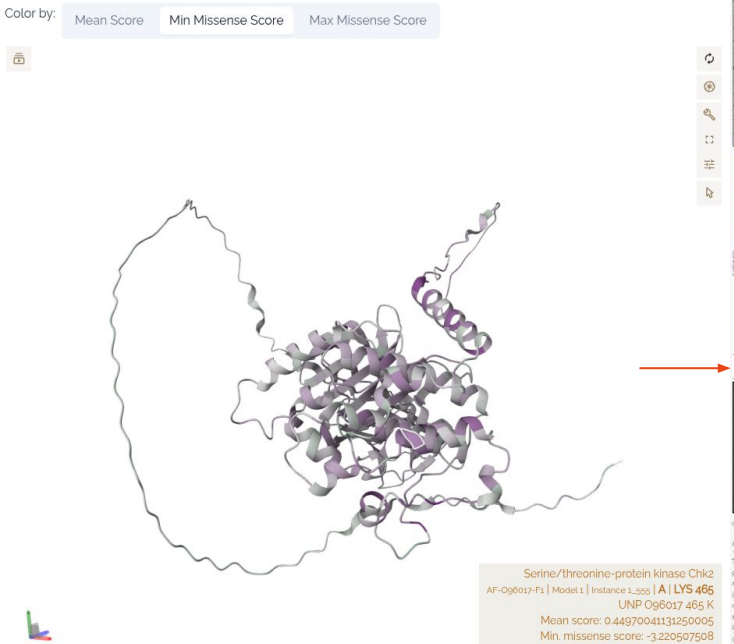
Thresholds: Gebbia M et al. (2024) — Method: Brnich SE et al. (2019)

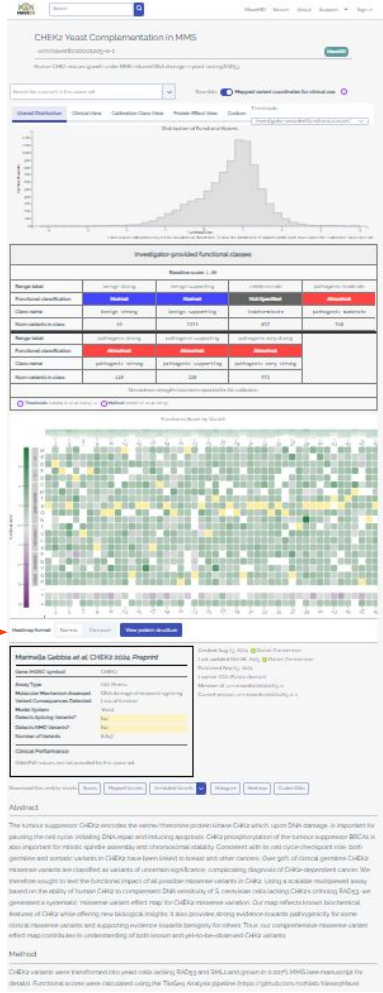


3 MAVEDB. CHK2: VE map



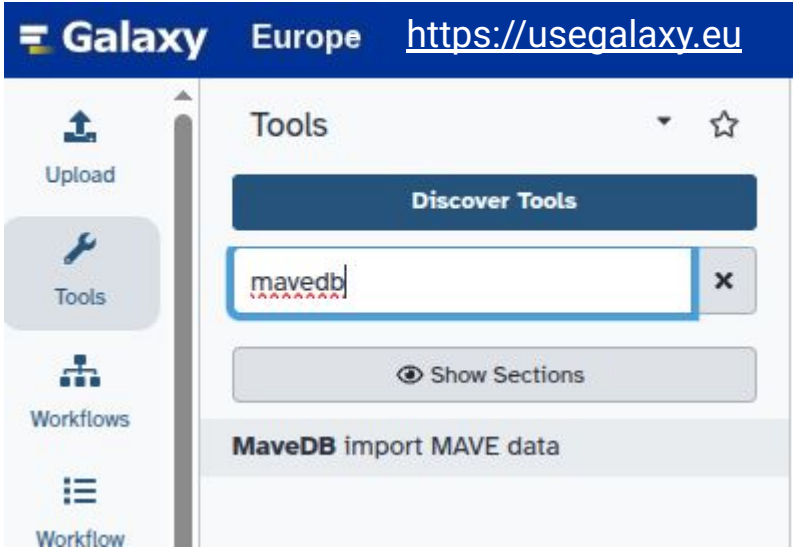
3 MAVEDB. CHK2: Structural mapping





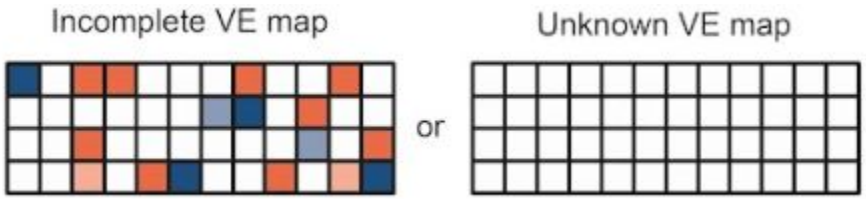
3 MAVEDB. Practical integration into bioinformatics workflows

- Reuse/integrate datasets
  - API
  - Web
  - **Galaxy tool**
- Inspect metadata (assay type, reference sequence)
- **Normalize** scores
- Use for downstream study



4 SPARSITY PROBLEM. Real-world DMS datasets are sparse

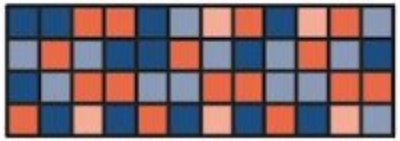
- Low sequencing depth
- Experimental bias
- Etc.



4 SPARSITY PROBLEM. Sparsity matters

- Hard to compare across studies
- Incomplete clinical interpretation
- Structural analyses benefit from complete maps
- ML requires dense matrices

Complete VE map



5 IMPUTATION. Supervised learning

Input: observed variant effects

Features:

- Sequence embeddings
- Evolutionary conservation
- Physicochemical descriptors

Output: predicted missing scores

5 IMPUTATION. Existing approaches

- Conservation-based models (SIFT, PROVEAN)
- ML-based
  - Random forests (Mighell et al., 2018; Weile et al., 2017)
  - Gradient boosting (Envision, VEFill)
  - Neural networks (Deep2Full)
- Hybrid models (CPT-1)

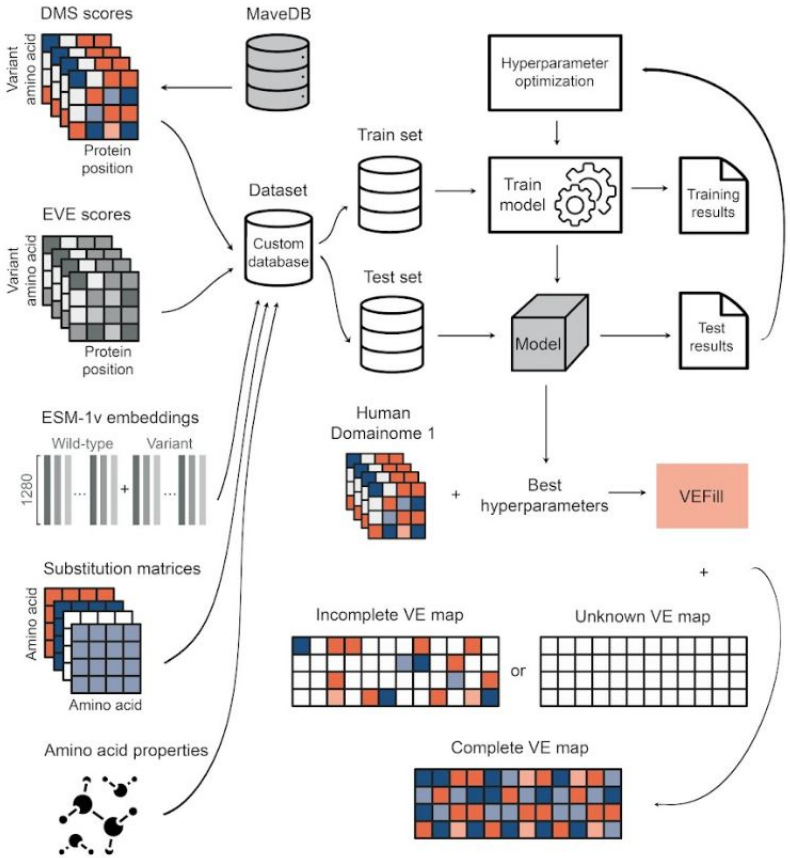
6 VEFILL. Variant Effect Imputation Model

Designed for incomplete DMS datasets

Integrates:

- Protein language model embeddings (ESM-1v)
- Evolutionary features (EVE, substitution matrices, e.g. BLOSUM62)
- Biochemical properties (charge, hydrophobicity, etc.)

Outputs predicted effect scores



6 VEFILL. Feature design

PLM embeddings → contextual representation

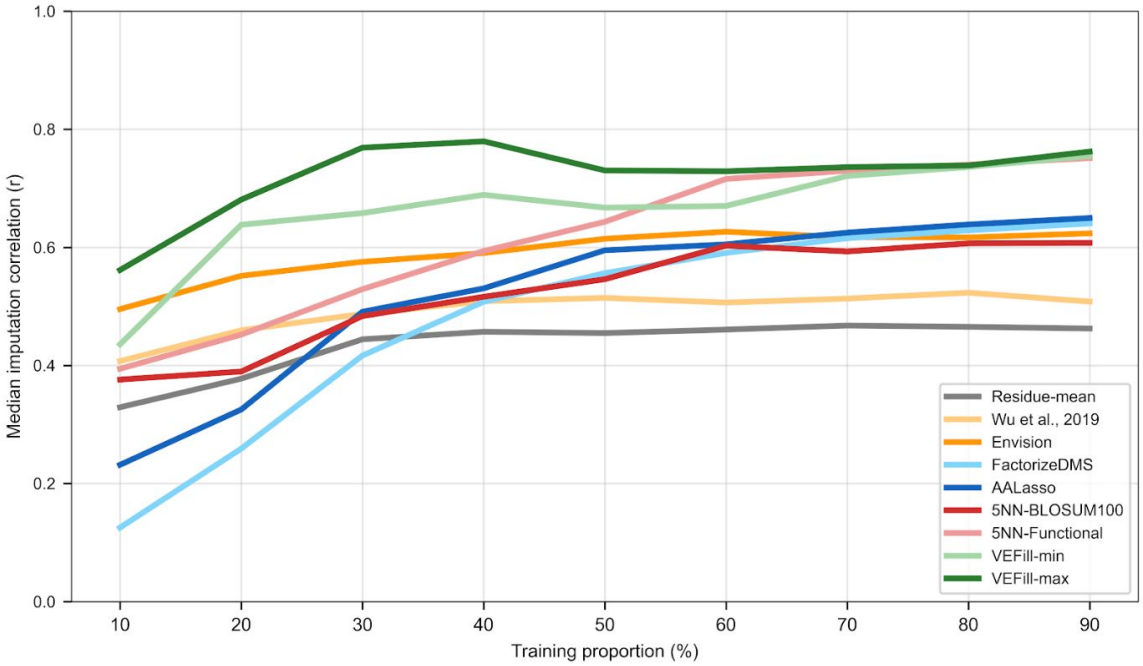
Evolution → selective constraint

Biochemistry → structural plausibility

6 VEFILL. Performance

- Performance remains strong even at low training fractions
- Results based on random downsampling
- Does not guarantee recovery of all clinically important variants

- Future work:
  - Systematic evaluation of targeted low-coverage DMS designs



6 VEFILL. End-to-end workflow with VEFill

1. Retrieve dataset from MaveDB
2. Preprocess and normalize scores
3. Train VEFill
4. Impute missing variants
5. Interpret
6. Integrate into downstream studies

VEFill preprint
(paper coming soon)

Responsible use of DMS data

- Computational predictions ≠ functional assay results
- Functional assay results ≠ clinical truth

- Assay-specific context matters
- Avoid direct clinical overinterpretation
- Follow
  - ACMG/AMP guidelines (Richards et al., 2015)
  - Recommendations (Brnich et al., 2019)

DMS ≠ clinical truth

Future directions

- Cross-assay harmonization
- Multi-protein generalization
- Integration with clinical variant databases (e.g., gnomAD)
- Community standards
- Toward a complete Variant Effect Atlas

Summary

- DMS provides scalable functional evidence
- MaveDB enables open reuse
- Real datasets are sparse
- ML-based imputation fills gaps
- VEFill demonstrates integrative approach
- Responsible application is essential

Thank you for your attention!

Acknowledgements

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Australian BioCommons

Galaxy Europe Team

My contacts

Matrix: @polina.polunina:matrix.org

Email: polunina@informatik.uni-freiburg.de

GitHub: github.com/PlushZ

References

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